

Day 4 — 3-Digit $\times$ 11

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute $234 \times 11 = 2_5_7_4$ using the extended split method.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. ““by one more than the previous” — sūtra 1 in Tirthajī”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Generalise to 3-digit.** For abc ($= 100a + 10b + c$) $\times 11$: $(100a + 10b + c)(10 + 1) = 1000a + 100b + 10c + 100a + 10b + c = 1000a + 100(a+b) + 10(b+c) + c$
2. **Procedure.** Split abc with two gaps: **a _ b _ c**. Fill gaps with adjacent sums: $a+b$, then $b+c$. Result: **a (a+b) (b+c) c**. Carry where needed.
3. **Worked examples.**
 - $234 \times 11 = 2_ (2+3)_ (3+4)_ 4 = 2574$
 - $567 \times 11 = 5_ (5+6)_ (6+7)_ 7 = 6237$ (carries: $11 \rightarrow$ carry 1 to left; $13 \rightarrow$ carry 1 to next left)
 - $729 \times 11 = 7_ (7+2)_ (2+9)_ 9 = 8019$
4. **Algebra check.** $234 = 100 \cdot 2 + 10 \cdot 3 + 4$. $\times 11 = 1000 \cdot 2 + 100 \cdot 5 + 10 \cdot 7 + 4 = 2574$. ✓
5. **Extension activity.** Each pair solves five 3-digit $\times 11$ problems, then verifies by long multiplication. Time-budget 8 minutes; share fastest correct-answer time.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces $3\text{-Digit} \times 11$.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).